

Perdiction Results

2024-10-17

Bill Passage

- **Number of observations:** 40,000.

Table 1: True and GPT-4o Predicted Bill Passage Rates

Model	Passage Rate
<i>House</i>	
True	0.074
GPT-4o w/o date restriction	0.043
GPT-4o w/ date restriction	0.384
<i>Senate</i>	
True	0.060
GPT-4o w/o date restriction	0.042
GPT-4o w/ date restriction	0.330

Table 2: Accuracy, TPR, and FPR of GPT-4o Predicted Bill Passage Rates

Model	Accuracy	TPR	FPR
<i>House</i>			
GPT-4o w/o date restriction	0.912	0.198	0.031
GPT-4o w/ date restriction	0.644	0.691	0.359
<i>Senate</i>			
GPT-4o w/o date restriction	0.925	0.225	0.031
GPT-4o w/ date restriction	0.695	0.711	0.306

Figure 1: Accuracy of Bill Passage Predictions, GPT-4o

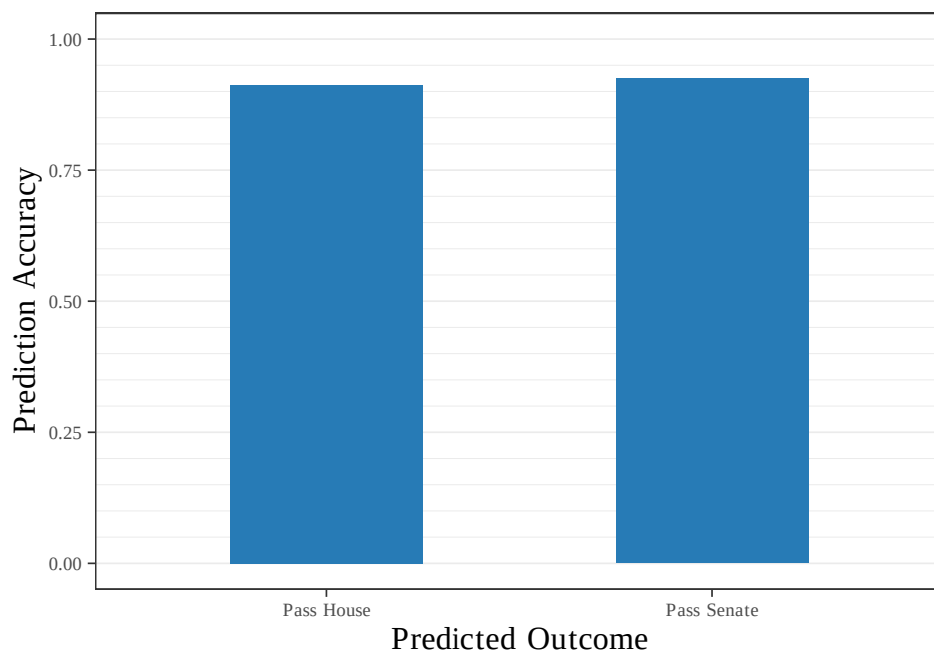


Figure 2: Accuracy of Bill Passage Predictions, GPT-4o, by Prompt

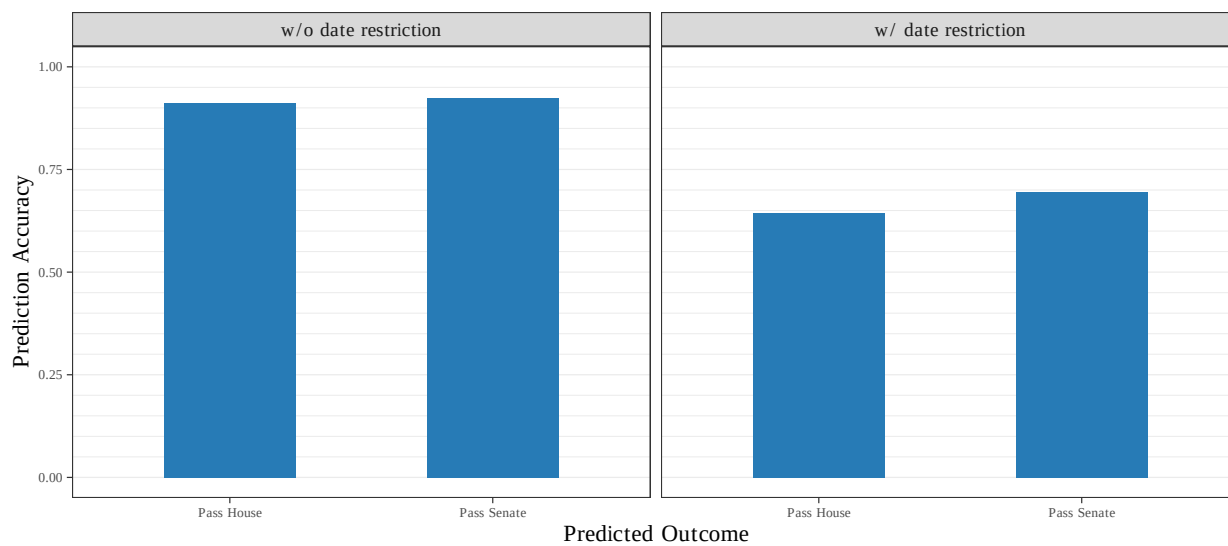


Figure 3: TPR of Bill Passage Predictions, GPT-4o, by Prompt

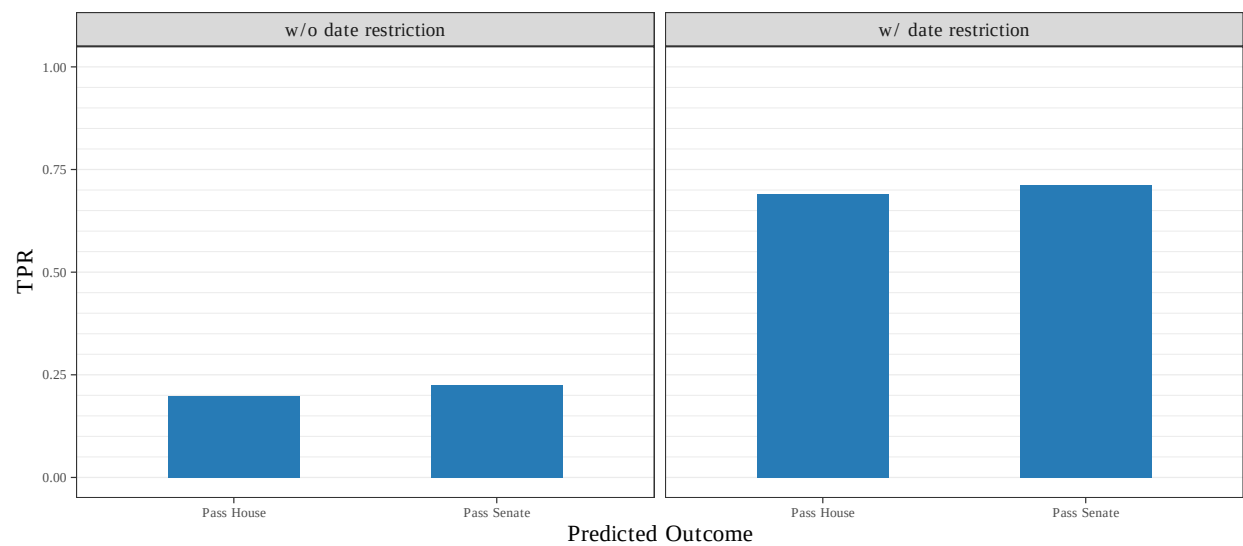
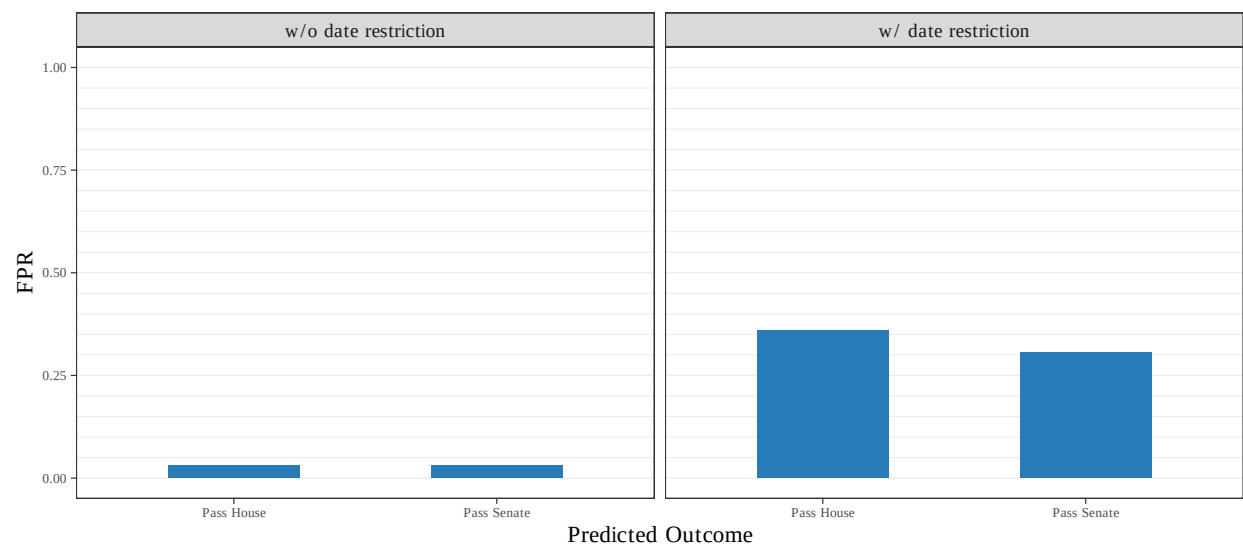


Figure 4: FPR of Bill Passage Predictions, GPT-4o, by Prompt



Bill Summary Completion

- **Number of Observations:** 40,000, with 39 responses exceeding the 750-token limit.
- **Cleaning:** For responses missing the provided part of the bill summary, we append it. If a response starts differently from the provided summary, we leave it as is. Both the true and LLM summaries are cleaned by removing **non-alphanumeric** characters and converting everything to lowercase.
- **Embedding Model:** Cleaned summaries are embedded using OpenAI’s **text-embedding-3-small** model, with an embedding size of 1536.

Examples of Successful Completion

Here are 10 examples where GPT-4o is considered to successfully complete the bill summary. Specifically, the cleaned responses are identical to the cleaned true summaries.

Table 3: Examples of Successful Completion of Bill Summaries (Non-Embedded)

Model	Prompt	BillID	DescriptionTrim	DescriptionClean	DescriptionLLMClean
GPT-4o	w/o date restric- tion	114-S- 1482	A bill to improve and reauthorize provisions relating to the applicat	a bill to improve and reauthorize provisions relating to the application of the antitrust laws to the award of needbased educational aid	a bill to improve and reauthorize provisions relating to the application of the antitrust laws to the award of needbased educational aid
GPT-4o	w/o date restric- tion	113-HR- 3616	To amend title XVIII of the Social Security Act to distribute additional informati	to amend title xviii of the social security act to distribute additional information to medicare beneficiaries to prevent health care fraud and for other purposes	to amend title xviii of the social security act to distribute additional information to medicare beneficiaries to prevent health care fraud and for other purposes

Table 3: Examples of Successful Completion of Bill Summaries (Non-Embedded) (*continued*)

Model	Prompt	BillID	DescriptionTrim	DescriptionClean	DescriptionLLMClean
GPT-4o	w/o date restriction	106-S-2700	A bill to amend the Comprehensive Environmental Response, Compensation, and Liability Act of 1980 to promote the cleanup and reuse of b	a bill to amend the comprehensive environmental response compensation and liability act of 1980 to promote the cleanup and reuse of brownfields to provide financial assistance for brownfields revitalization to enhance state response programs and for other purposes	a bill to amend the comprehensive environmental response compensation and liability act of 1980 to promote the cleanup and reuse of brownfields to provide financial assistance for brownfields revitalization to enhance state response programs and for other purposes
GPT-4o	w/o date restriction	107-S-39	A bill to provide a national medal for public safety officers who act with e	a bill to provide a national medal for public safety officers who act with extraordinary valor above and beyond the call of duty and for other purposes	a bill to provide a national medal for public safety officers who act with extraordinary valor above and beyond the call of duty and for other purposes
GPT-4o	w/o date restriction	113-S-1799	A bill to reauthorize subtitle A of t	a bill to reauthorize subtitle a of the victims of child abuse act of 1990	a bill to reauthorize subtitle a of the victims of child abuse act of 1990
GPT-4o	w/ date restriction	105-S-2110	A bill to authorize the Federal programs to preve	a bill to authorize the federal programs to prevent violence against women and for other purposes	a bill to authorize the federal programs to prevent violence against women and for other purposes
GPT-4o	w/ date restriction	102-HR-5431	To designate the Federal building located at 200 Federal Plaza	to designate the federal building located at 200 federal plaza in paterson new jersey as the robert a roe federal building	to designate the federal building located at 200 federal plaza in paterson new jersey as the robert a roe federal building

Table 3: Examples of Successful Completion of Bill Summaries (Non-Embedded) (*continued*)

Model	Prompt	BillID	DescriptionTrim	DescriptionClean	DescriptionLLMClean
GPT-4o	w/ date restriction	105-S-621	A bill to repeal the Public Utility Holding Company Act of 1935, to enac	a bill to repeal the public utility holding company act of 1935 to enact the public utility holding company act of 1997 and for other purposes	a bill to repeal the public utility holding company act of 1935 to enact the public utility holding company act of 1997 and for other purposes
GPT-4o	w/ date restriction	113-S-42	A bill to provide anti-retaliation pro	a bill to provide antiretaliation protections for antitrust whistleblowers	a bill to provide antiretaliation protections for antitrust whistleblowers
GPT-4o	w/ date restriction	106-S-453	A bill to designate the Federal building located at 709 West 9th S	a bill to designate the federal building located at 709 west 9th street in juneau alaska as the hurff a saunders federal building	a bill to designate the federal building located at 709 west 9th street in juneau alaska as the hurff a saunders federal building

Tables and Figures

Table 4: Share of Identical True and GPT-4o Bill Summaries in Non-Embedded Form

Model	Average
GPT-4o w/o date restriction	0.034
GPT-4o w/ date restriction	0.033

Table 5: Cosine Similarity and Euclidean Distance Between True and GPT-4o Embedded Bill Summaries

Model	Average	Baseline
<i>Cosine Similarity</i>		
GPT-4o w/o date restriction	0.830	0.379
GPT-4o w/ date restriction	0.829	0.379
<i>Euclidean Distance</i>		
GPT-4o w/o date restriction	0.536	1.110
GPT-4o w/ date restriction	0.538	1.110

Figure 5: Cosine Similarity Between True and LLM Bill Summaries Compared to a Random Baseline, GPT-4o, by Prompt

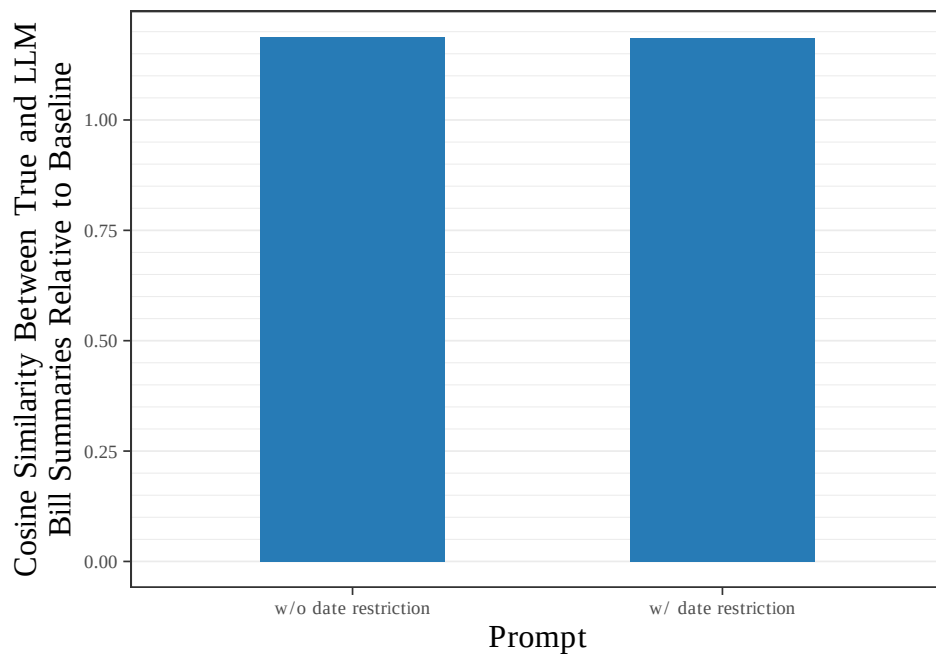


Figure 6: Euclidean Distance Between True and LLM Bill Summaries Compared to a Random Baseline, GPT-4o, by Prompt

